Hongliang Li

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Profile

AI4science researcher specializing in biological foundation models and drug discovery. Experienced in building unified biology foundation models (protein design, dynamic modeling, affinity prediction) and drug discovery workbenches integrating knowledge across disciplines and continuously supporting analysis, design, validation, and decision-making throughout the drug discovery process.

Education

- **Fudan University — Research Institute of Intelligent Complex Systems** *Starting Aug 2026*
  Ph.D. Student | Biology Foundation Models & AI-Driven Drug Discovery Shanghai, China
  - Advisors: Prof. Shuangjia Zheng and Prof. Siqi Sun
- **Beijing Institute of Technology** *Sep 2020 – Jun 2023*
  M.S. in Computer Science | NLP & Biological Language Models Beijing, China
  - Advisor: Prof. Bin Liu
- **Beijing Institute of Technology** *Sep 2016 – Jun 2020*
  B.S. in Mechatronic Engineering Beijing, China

Professional Experience

- **AI Department, Lingang Laboratory** *Jul 2025 – Jul 2026*
  Research Engineer Shanghai, China
  - Developed and optimized unified affinity-scoring models spanning small molecules, antibodies, and peptides, with multi-target binding prediction capabilities.
  - Built an AI-enabled drug-design workbench for structure–function studies and end-to-end candidate profiling; implemented agent-based provenance tracking to improve reproducibility and computational validation across multi-step workflows.
- **Institute of AI Industry Research, Tsinghua University** *Sep 2023 – Apr 2025*
  Research Assistant, ATOM Lab Beijing, China
  - Contributed to a Ministry of Science and Technology initiative on deep-learning-based end-to-end protein structure prediction. Advanced neural-network methods for protein conformation generation and structural validation.

Selected Projects

- **AIRFold — Systematic Protein Structure Prediction** Homepage | GitHub
  - Developed at the **Institute for AI Industry Research (AIR), Tsinghua University**, within the ATOM Lab led by **Prof. Yanyan Lan**; contributed to the National Key R&D initiative on deep-learning-based end-to-end protein structure prediction.
  - Helped build a scalable, systematic prediction platform combining homology mining and MSA generation with AlphaFold2, RoseTTAFold2, OmegaFold, ESMFold, and model-quality estimation in a modular microservice architecture.
  - Advanced protein conformation generation, structural validation, and model ranking. The resulting system achieved top-ranking performance in the continuous automated model evaluation (CAMEO) benchmark.
- **OriDyn — All-Modality Conformational Ensemble Generation** In Development
  - Developing an AlphaFold 3-based framework for generating conformational ensembles across biomolecular modalities, including proteins, protein–ligand complexes, antibodies, peptides, and other molecular assemblies.
  - Designed ensemble sampling, structural filtering, and confidence-based ranking strategies to move beyond single-structure prediction and recover diverse, biologically relevant conformational states.
  - On current internal benchmarks, OriDyn substantially outperforms published conformational-ensemble baselines, including BioEmu, across the evaluation settings used; broader external validation is ongoing.
  - Enables dynamics-aware analysis for binding-site discovery, affinity modeling, molecular design, and downstream drug-discovery workflows.

- **ScoreBench — All-Modality Affinity Prediction Benchmark** Benchmark & Evaluation
  - Constructed a comprehensive affinity-prediction benchmark for all-modality drug systems, spanning small molecules, antibodies, peptides, and other therapeutic modalities.
  - Organized the benchmark around three practical task families: **binding** assessment, affinity **optimization**, and mutation-effect prediction, with standardized data, evaluation protocols, and metrics for consistent comparison.
  - Conducted broad evaluations of frontier large language models, including **Claude, GPT, Kimi, and DeepSeek**, to characterize their capabilities across drug modalities and affinity-related tasks.
  - Performed task- and modality-level error analysis to identify recurring LLM failure modes, robustness gaps, and limitations in molecular reasoning and quantitative affinity prediction.
- **OriBench — OriGene AI Workbench** Platform | Documentation
  - Built an AI-powered scientific workbench integrating machine-learning models, molecular modeling tools, and multi-omics data for rational drug design and hit-to-lead optimization.
  - Connected analysis, design, validation, and decision-making in reproducible multi-step workflows, with agent-based provenance tracking for computational results and design rationale.

Publications

Total citations: ***404***

Submitted / Under Review

- **Hongliang Li** et al. (2026). "OriScore: Unified Affinity Scoring Framework for All Drug Modalities." *Nature* (under review).
  A unified affinity-scoring framework spanning small molecules, antibodies, and peptides, with multi-target binding prediction capabilities. ***Advised by Prof. Bowen Zhou.***

Peer-Reviewed

- **Hongliang Li** and Bin Liu. (2023). "BioSeq-Diabolo: Biological Sequence Similarity Analysis Using Diabolo Embeddings." *PLOS Computational Biology*, 19(6), e1011214.
- **Hongliang Li**, Yihe Pang, and Bin Liu. (2022). "BioSeq-BLM: A Comprehensive Platform for Analyzing DNA, RNA, and Protein Sequences Based on Biological Language Models." *Nucleic Acids Research*, 49(22), e129.
- Qilong Wang, Huikun Zeng, Yan Zhu, **Hongliang Li**, et al. (2021). "Dual UMIs and Dual Barcodes With Minimal PCR Amplification Removes Artifacts and Acquires Accurate Antibody Repertoires." *Frontiers in Immunology*, 12, 778298.
- **Hongliang Li**, Hongyu Wang, Dongping Chen, and Zhengshe Kang. (2020). "Revealing the Optical Properties of Polycyclic Aromatic Hydrocarbon Clusters with Surface Formyl Groups." *Proceedings of the Combustion Institute*, 38(1), 1207–1215.
- **Hongliang Li**, Zhengshe Kang, and Dongping Chen. (2019). "HOMO–LUMO Gaps of Polycyclic Aromatic Hydrocarbon Clusters with Formyl Groups." *12th Asia-Pacific Conference on Combustion (ASPACC 2019)*.

Research Expertise

- **Biological Foundation Models:** multimodal biomolecular representation learning, protein structure and conformational-ensemble generation, and structure–function modeling.
- **AI-Driven Drug Discovery:** unified affinity prediction across drug modalities, molecular design, hit-to-lead optimization, and multi-target modeling.
- **Scientific AI Systems:** scalable model orchestration, agent-based scientific workflows, provenance tracking, computational validation, and reproducible decision support.